DSA VISUALISED · EP 01

@scicomppaths

Binary Search — cheat sheet

Sorted data · half the range gone every step · O(log n)

```
// template (Java) - memorise this
int lo = 0, hi = a.length - 1;
while (lo <= hi) {
    int mid = lo + (hi - lo) / 2;   // no overflow
    if (a[mid] == target) return mid;
    if (a[mid] < target) lo = mid + 1;  // right half
    else                 hi = mid - 1;  // left half
}
return -1;  // not found
```

```
// how many steps? (worst case)
```

16	→ 4 steps (reel example: found 42 in 3)
1 crore	→ 24 steps (linear: 1 crore steps)
100 crore	→ 30 steps
Time / Space	O(log n) / O(1)

```
// the overflow bug - an interview favourite
```

Java int max ≈ 214 crore (2,147,483,647).
low = 150 cr, high = 200 cr → low + high = 350 cr → overflow → negative → crash.
Fix: mid = low + (high − low) / 2 = 150 + 25 = 175 cr ✓
This bug sat in Java's own Arrays.binarySearch for ~9 years (fixed in 2006).

```
// variants
```

First occurrence	on a match: ans = mid, hi = mid − 1 (keep searching left)
Last occurrence	on a match: ans = mid, lo = mid + 1
Lower bound	first index where a[i] ≥ target
On answer	search the range of answers: ok(mid) ? hi = mid : lo = mid + 1
Rotated array	one half is always sorted — use it to decide

```
// practice (LeetCode)
```

704	Binary Search
35	Search Insert Position
34	First and Last Position of Element
33	Search in Rotated Sorted Array
153	Find Minimum in Rotated Sorted Array
875	Koko Eating Bananas (on answer)
1011	Capacity To Ship Packages (on answer)

```
// real world
```

git bisect: finds the bad commit among 1000 in ~10 steps. Database indexes (B-trees) use the same idea.
Built-in: Java Arrays.binarySearch · C++ lower_bound · Python bisect

Save it · for quick revision

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